```
  int bounce1 = digitalRead(switchPin);
  delay(25);
  int bounce2 = digitalRead(switchPin);

  while ((bounce1 == bounce2) && (bounce1 == LOW)) {
    oldPin = newPin;
    newPin++;

    if (newPin == 3) newPin = 0;

    digitalWrite(rgb[oldPin], HIGH);
    delay(time);
    digitalWrite(rgb[newPin], HIGH);
    delay(time);
    digitalWrite(rgb[oldPin], LOW);

    if (newPin == 0) {
      for (int i=0; i<3; i++) digitalWrite(rgb[i], HIGH);
      delay(time);
      for (int i=0; i<3; i++) digitalWrite(rgb[i], LOW);
    }
    bounce1 = digitalRead(switchPin);
    delay(25);
    bounce2 = digitalRead(switchPin);
  }
  for (int i=0; i<3; i++) digitalWrite(rgb[i], LOW);
  delay(25);
}
```

Source Code Summary

This sketch looks dramatically different from our last one, so let's take it one step at a time to see how it works beginning our summary at the top with variable declarations. We start with the following three, constant, global variables that we will use throughout our sketch:

```
const int rgb[] = {9,10,11};
const int time = 250;
const int switchPin = 2;
```

The variable `rgb[]` is actually a special type of variable called an array that will be explained fully in Chapter 8. It contains the three pin numbers for our RGB LED (9, 10, 11) and each position in the array can be accessed by an index number (0, 1, 2). The variable `time` is used for the delay between each color—we speed it up here to make it more interesting. Finally `switchPin` is the pin that our tilt switch is connected to.

We use our `setup()` function to set the mode for each of the I/O pins that we are using, as follows:

```
for (int i=0; i<3; i++) pinMode(rgb[i], OUTPUT);
pinMode(switchPin, INPUT);
```

The first line uses a little trick with a `for` loop to step through each index of the array to configure each pin as an output. The next line establishes the input pin for our switch.